



DO-IT-YOURSELF: PROJECT

PARTS LIST

Semiconductors:

Board1	- Arduino Uno board
MOD1	- 5V, single changeover (1CO) relay module
MOD2	- MPU6050 module
MOD3	- Mercury tilt switch module

Miscellaneous:

S1	- On/off switch (6A)
	- 12V, 1A DC power supply
	- Electric iron
	- Jumper wires
	- 2.1mm DC power jack
	- 3-pin power socket

Uno, 12V SMPS power supply, and the relay module inside the plastic box. The MPU6050 module and mercury tilt switch module should be placed inside the iron box at appropriate orientations to sense the vertical and horizontal positions of the iron box.

Use a 6-core soft cable to connect (common ground and common 5V, SDA, SCL, INT pins of MPU6050 module, signal out pin of tilt switch

module) to the Arduino Uno module, which is enclosed in the plastic box. The relay remains switched on as long as the iron box is in motion, but it switches off when the iron box is placed in vertical position.

Use bypass switch S1 to disconnect the circuit when smart switch action is not needed. Fig. 9 shows the proposed construction details of the project.

Precautions

The Arduino Uno board is highly sensitive. Handle it carefully. Check the DC power supply jack input polarity (2.1mm centre-positive plug into the board's power jack). Take care while placing the 12V SMPS module and the Arduino Uno board into the plastic switch box. Use of a 1000W iron box is recommended.

Choose an appropriate place inside or outside the iron box to place

EFY Note

The source code of this project is available for free download at source.efymag.com

the sensors. The position of the tilt switch is very important. Place the tilt switch in such a way that it can detect vertical and horizontal positions of the iron box easily. **EFY**



K. Murali Krishna works as junior engineer at BSNL, Rajahmundry, Andhra Pradesh. He is an electronics enthusiast, circuit designer, and technical writer

Job Guaranteed.

You don't need a training institute. Just do it yourself.

WORK WITH OPEN SOURCE.

www.opensourceforu.com

Demand for Open Source is sky rocketing. Be it for managing IT infrastructure or development of software —Open Source experts are what organisations are seeking. Develop expertise in an Open Source stack, and watch your career graph zoom.

You don't need a training institute for this. You can do it yourself. And, *Open Source For You* can be your friend and a guide through this journey.

TO READ OUR PRINT EDITION Visit: <https://subscribe.efyindia.com>

TO READ OUR EZINE EDITION Visit: <https://ezine.lfyimag.com>

WORLD'S LEADING PUBLICATION ON OPEN SOURCE

Looking for marketing solutions to engage with cutting edge techies?
Contact us at growmybiz@efy.in OR call us at +91-9811155335.

